

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2, BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

I **M Nishanth (192511154)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M Nishanth*

Annexure A

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.
3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.
5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication

between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

SIMATS ENGINEERING

ASSESSMENT TOOL 2 – CSA0706 COMPUTER NETWORKS (CO4)

Problem-Solving Methodology Used Across All Five Cases

Each of the five industry scenarios below is worked through the same structured, five-step methodology, so the approach is systematic and repeatable rather than ad hoc:

1. Requirement Analysis – identify the organization's scale, budget, uptime/security needs, and growth expectations.
2. Topology Selection – match the requirement profile to the most suitable topology (star / mesh / bus / hybrid), justified against the alternatives.
3. Device & Technology Selection – choose specific, real-world hardware, cabling standards, and protocols appropriate to the chosen topology and budget.
4. Addressing & Implementation Design – define IP addressing/VLAN scheme and how the design is physically and logically realized.
5. Testing & Validation – define concrete checks that would confirm the design meets its requirements before go-live.

This same structure is applied to every scenario below, ensuring consistent documentation and making the reasoning behind each design decision explicit and auditable – the kind of repeatable process a real network engineering team would follow, and one that would scale naturally to a team-based project where different members own different steps (e.g., one member handles requirement/stakeholder interviews, another handles device selection and costing, another handles addressing design and validation testing).

1. Startup Office Network (20 Users) – Star Topology

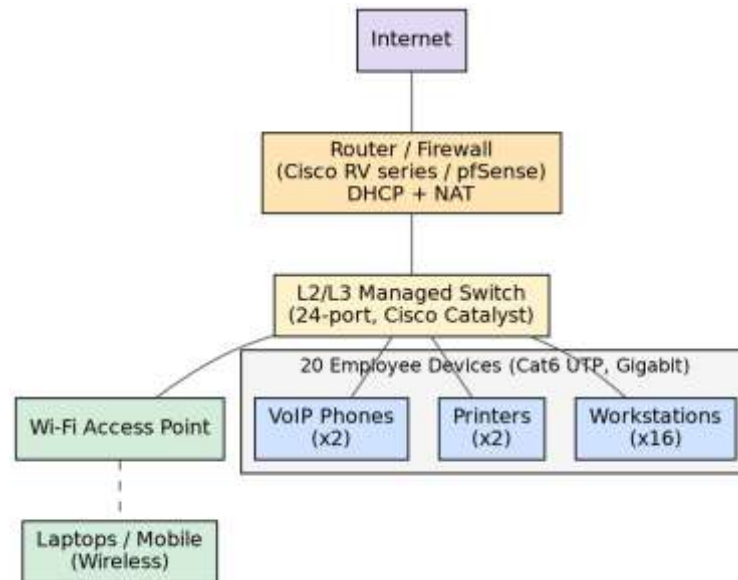
Given (Original Statement): A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

1.1 Problem Analysis

Key Requirement: A simple, affordable, easily-manageable network for 20 users with no dedicated IT department, that isolates faults and leaves room to grow.

1. Small user base (20 people) means the network doesn't need enterprise-scale redundancy – cost-effectiveness and ease of management matter more than fault-tolerant core design.
2. Mixed device types (workstations, printers, VoIP phones, wireless laptops) must all be supported from one manageable core.
3. No dedicated network engineer is implied, so troubleshooting must be simple enough for general IT staff – favoring a design where a fault is easy to isolate to one cable or device.
4. Growth headroom is needed without a full redesign, since startups scale quickly.

1.2 Solution Design



A star topology directly satisfies every requirement above: the central managed switch gives one clear point of management and troubleshooting, fault isolation means a single bad cable never takes down the office, and Cat6 UTP at Gigabit speeds comfortably covers a typical office floor plan (100 m max run per TIA/EIA-568 standard) without any specialized cabling.

1.3 Technical Implementation

Component	Specification	Purpose
Router/Firewall	Cisco RV345 or pfSense appliance	Internet gateway, DHCP server, NAT, basic firewall/VPN
Core Switch	24-port L2/L3 managed switch (Cisco Catalyst 1000 series / Netgear ProSafe)	Central connectivity hub; supports future VLAN segmentation
Wireless AP	Ubiquiti UniFi / Cisco Business AP (PoE-powered)	Wireless coverage for laptops and mobile devices
Cabling	Cat6 UTP, max 100 m runs (TIA/EIA-568-B standard)	Gigabit wired connectivity to every device
IP Addressing	192.168.1.0/24; DHCP pool .100–.200; static reservations for printers/VoIP (.10–.20)	Simple, single-subnet addressing appropriate for 20 users

1.4 Testing & Validation

- Certify every Cat6 run with a cable tester (e.g., Fluke LinkRunner) to confirm Gigabit-standard wiring before deployment.
- Run ping/traceroute from each workstation to the internet gateway to confirm DHCP and NAT are functioning correctly.
- Fault-isolation drill: physically disconnect one workstation's cable and confirm the other 19 devices remain fully connected.
- Load test: simulate simultaneous large file transfers plus an active VoIP call to confirm the 24-port switch and uplink handle combined traffic without noticeable call-quality degradation.
- Conduct a basic wireless site survey to confirm the single AP provides adequate signal strength across the whole office floor plan.

1.5 Innovation / Enhancement Beyond the Base Design

- Use a PoE (Power over Ethernet) switch so the Wi-Fi AP and VoIP phones draw power directly over their Ethernet cable, eliminating separate power adapters and reducing cable clutter.
- Even at 20 users, pre-configure a separate VLAN for VoIP traffic with basic QoS prioritization – a low-cost, forward-looking step that protects call quality as data traffic grows, without requiring any topology change later.

2. Bank Data Center Network – Partial Mesh Topology

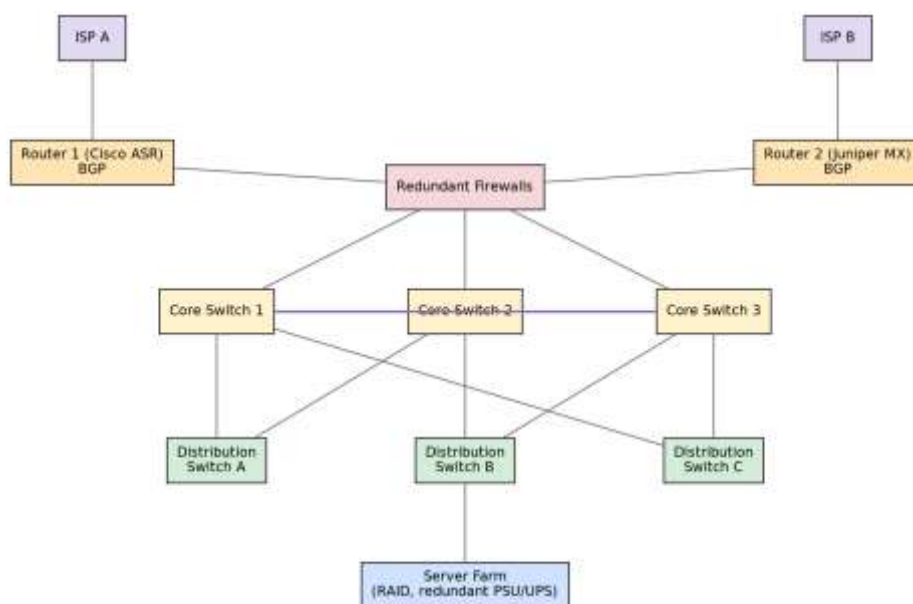
Given (Original Statement): A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

2.1 Problem Analysis

Key Requirement: Near-zero downtime and low-latency, fault-tolerant infrastructure for financial transactions, meeting regulatory uptime and data-integrity requirements.

- Financial transactions cannot tolerate outages – even brief downtime has direct monetary and regulatory-compliance consequences.
- Redundancy is needed at every layer: links, core devices, internet connectivity, power, and storage – a single point of failure anywhere is unacceptable.
- Latency must remain low and predictable for real-time transaction processing, ruling out transmission media or topologies with unpredictable delay.
- The solution must support fast automatic failover (milliseconds, not minutes) since manual intervention is too slow for financial-grade SLAs.

2.2 Solution Design



A partial mesh at the core (rather than a full mesh everywhere, which would be prohibitively expensive to cable at scale) gives every core switch multiple direct paths to every other core switch, while distribution switches keep costs reasonable by uplinking to just two core switches each – still eliminating any single point of failure on the path from an access device to the core.

2.3 Technical Implementation

Component	Specification	Purpose
Core Switches	3+ interconnected core switches (partial mesh)	No single core device/link failure can isolate the network
Routers	Dual Cisco ASR1001-X / Juniper MX204, BGP multihomed to 2 ISPs	Redundant internet connectivity with automatic ISP failover

Firewalls	Redundant HA pair (e.g., Palo Alto / Fortinet, active-passive or active-active)	Security enforcement without becoming a single point of failure
Core Cabling	Single-mode fiber (long/inter-building), multi-mode OM4 (intra-data-center)	EMI immunity, high bandwidth, low latency for financial transactions
Routing Protocols	OSPF (with BFD) internally; BGP externally	Sub-second automatic rerouting on link or ISP failure
Power & Storage	Dual UPS + generator backup; RAID 10 storage arrays	Protects against power loss and disk failure without service interruption

2.4 Testing & Validation

- Link-failure drill: physically disconnect one core-to-core fiber link and measure reroute convergence time (target: sub-second with OSPF + BFD).
- ISP-failure drill: simulate a primary ISP outage and confirm BGP fails over to the secondary ISP with minimal to no dropped transaction sessions.
- Firewall failover test: fail the active firewall in the HA pair and confirm the passive unit takes over transparently.
- Peak-load testing during simulated high-volume periods (e.g., month-end batch processing) to confirm bandwidth headroom on all mesh links.
- Compliance audit of the design against the bank's regulatory uptime SLA (e.g., 99.999%) and data-integrity requirements.

2.5 Innovation / Enhancement Beyond the Base Design

- Layer an SD-WAN overlay on top of the dual-ISP BGP setup to dynamically select the lowest-latency/jitter path in real time, rather than relying on BGP path selection alone.
- Institute periodic, scheduled 'failure drills' (chaos-engineering style controlled outages) as an ongoing validation practice, not just a one-time test, to continuously verify the mesh's real-world failover behavior.

3. Classroom Lab Network (Tight Budget) – Bus Topology

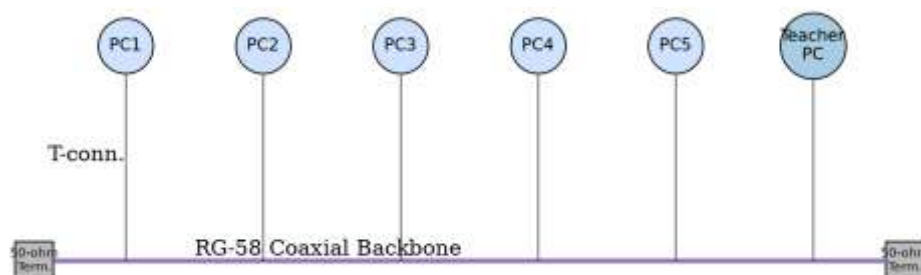
Given (Original Statement): For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

3.1 Problem Analysis

Key Requirement: The lowest possible hardware cost for a small, supervised, light-usage classroom lab, where enterprise-grade reliability is not required.

- Budget is the dominant constraint – minimizing hardware (no central switch/hub required in the classic design) directly minimizes cost.
- Usage is light and predictable (browsing, file sharing), so the classic bus limitation – performance degradation under heavy/collision-prone traffic – is not a practical concern.
- The environment is supervised (a teacher present), so the risk of an accidental single-cable-break outage is an acceptable, low-consequence trade-off.
- No dedicated server budget exists, so the design must rely on built-in OS file-sharing capability instead.

3.2 Solution Design



The classic coaxial bus (shown above) directly matches the original design brief and remains valuable to understand conceptually; in current practice, the modern equivalent – a single inexpensive unmanaged switch with Cat5e patch cables – achieves the same low cost and simplicity with hardware that is far easier to source and maintain today (see Section 3.5).

3.3 Technical Implementation

Component	Specification	Purpose
Backbone (classic)	RG-58 coaxial cable, 50Ω impedance	Single shared communication line for all PCs
Connectors (classic)	BNC T-connectors at each PC; 50Ω terminators at both cable ends	Taps each PC into the bus; prevents signal reflection
Backbone (modern variant)	8/16-port unmanaged switch + Cat5e UTP patch cables	Bus-like simplicity, minimal cost, easier to source/maintain
File Sharing	Teacher's PC using built-in Windows/Linux SMB or NFS sharing	No dedicated server needed for a light-usage lab
IP Addressing	Single flat subnet, e.g. 192.168.0.0/24, via a basic router or static IPs	Simple addressing appropriate for a small, single-room network

3.4 Testing & Validation

- Continuity-test the coaxial backbone and terminators with a cable tester to confirm no signal reflection issues before use.
- Verify file-sharing access from every student PC to the teacher's shared folder.
- Confirm adequate internet browsing bandwidth is available for the full class size simultaneously (particularly important if using the unmanaged-switch variant).
- Deliberately simulate a single cable break and document that the entire segment goes down as expected – confirming this known, accepted trade-off matches the design assumption rather than being an unexpected failure.

3.5 Innovation / Enhancement Beyond the Base Design

- Recommend the modern unmanaged-switch variant as the practical default for any new installation today, since true RG-58 coaxial hardware is largely obsolete and increasingly difficult to source – while still documenting the classic coax design for conceptual completeness, as done above.
- If the budget allows a small increment, suggest a basic managed switch instead of

unmanaged, enabling simple port-level diagnostics without materially increasing cost – a low-risk upgrade path for the future.

4. University Campus Network – Hybrid Topology

Given (Original Statement): A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

4.1 Problem Analysis

Key Requirement: A single, uniform topology cannot serve a multi-building campus with thousands of users and varied per-building needs – the design must deliver backbone redundancy, per-building manageability, and campus-wide scalability simultaneously.

- Scale (thousands of users, many buildings) rules out any flat, single-device-managed topology such as pure star or bus.
- Different building types have different priorities: labs need cost-efficiency, dorms need robust wireless density, admin buildings need reliability/security – a one-size-fits-all design can't optimize for all of them.
- A building-isolating outage (e.g., library losing all connectivity from one link failure) is unacceptable given the number of affected users, requiring backbone-level redundancy.
- The campus must be able to grow (new buildings, expanding labs) without redesigning or disrupting the existing network.

4.2 Solution Design

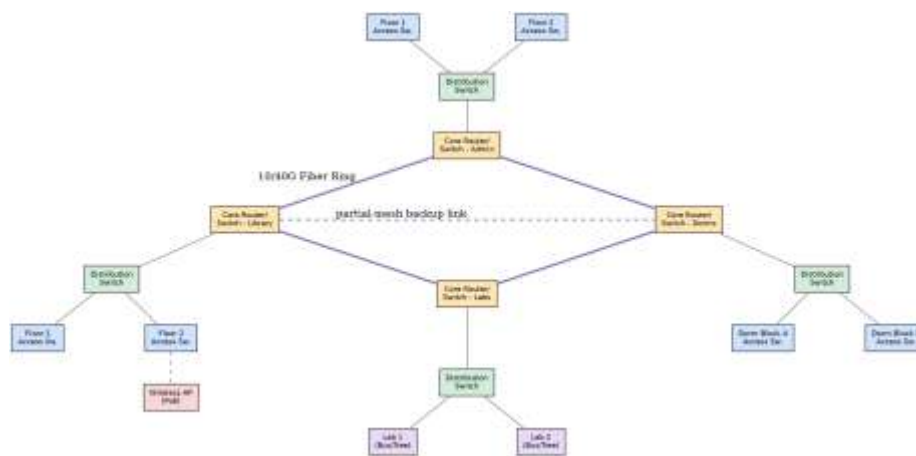


Figure 4.1: Three-Layer Hybrid Campus Design – Fiber Ring/Partial-Mesh Core, Per-Building Star Distribution/Access, PoE Wireless

The hierarchical three-layer hybrid – fiber ring/partial-mesh core, star-wired distribution-to-access within each building, and bus/tree structures for cost-sensitive labs – lets each layer be optimized independently: the core is optimized for redundancy, the building layer for manageability, and the lab layer for cost, exactly matching the differing requirements identified above.

4.3 Technical Implementation

Layer	Specification	Purpose
Core	Single-mode fiber ring/partial mesh, 10/40 Gbps; core routers & L3 multilayer switches (e.g., Cisco Catalyst 9500 / Juniper MX)	Backbone redundancy – no single link failure isolates a building
Distribution	Per-building L3 distribution switch with dual fiber uplinks to two core nodes	Building-level redundancy and traffic aggregation
Access	Floor-level L2 access switches (e.g., Cisco Catalyst 9200), Cat6 to end devices	Manageable, star-wired connectivity within each building
Wireless	Campus-wide APs (e.g., Cisco Meraki / Aruba) powered via PoE from access switches	Wireless coverage without separate power cabling
Addressing	Hierarchical IP scheme (e.g., 10.<building>.<floor>.0/24) enabling route summarization at the core	Efficient routing and simplified core routing tables as the campus grows

4.4 Testing & Validation

- Fiber-ring failure simulation: cut one ring segment and confirm automatic rerouting via the

remaining ring/mesh path with no building isolated.

- Cross-building latency and connectivity testing (e.g., dorm to library server) under both normal load and simulated peak load (e.g., semester registration period).
- Wireless roaming test – walk a test device between buildings and confirm seamless handoff between APs (validating 802.11r fast-roaming if implemented).
- New-building addition drill: simulate installing a new distribution switch and fiber run to the core, confirming zero disruption to existing building traffic during the change.

4.5 Innovation / Enhancement Beyond the Base Design

- Apply route summarization per building's addressing block at the core routers to keep core routing tables small and convergence fast even as the campus grows to many buildings.
- Introduce an SDN (Software-Defined Networking) controller for centralized, consistent policy management – QoS prioritization for classroom video streaming, and uniform security ACLs – applied campus-wide from a single control point rather than configured device-by-device.

5. Multi-Department Company Network – VLAN-Segmented Hierarchical Design

Given (Original Statement): Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

5.1 Problem Analysis

Key Requirement: Multiple departments sharing common infrastructure need logical separation for security and broadcast-domain control, while still being able to communicate with each other and the internet through a structured, scalable design.

- Departments (HR, Finance, Sales, IT) have different data-sensitivity levels – HR/Finance data should be logically isolated from general Sales traffic for security.
- A flat network would let broadcast traffic from any one department's devices flood every

other department, wasting bandwidth and increasing security exposure as the company grows.

- Departments still need controlled inter-department communication (e.g., IT must reach every department for support) and shared internet access.
- The design must scale cleanly as new departments or offices are added.

5.2 Solution Design

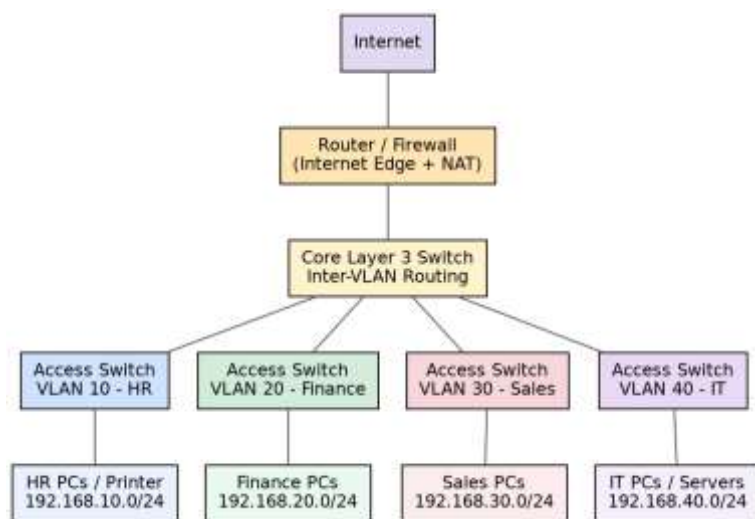


Figure 5.1: VLAN-Segmented Hierarchical Network – One VLAN per Department, Core L3 Inter-VLAN Routing

Assigning each department its own VLAN at the access layer, with a core Layer 3 switch performing inter-VLAN routing, gives exactly the isolation-plus-controlled-communication balance required: broadcast traffic stays contained within each department's VLAN, while the core switch's routing (optionally combined with ACLs) governs exactly which cross-department traffic is permitted.

5.3 Technical Implementation

VLAN ID	Department	Subnet	Access Switch Role
10	HR	192.168.10.0/24	L2 access switch, 802.1Q trunk to core
20	Finance	192.168.20.0/24	L2 access switch, 802.1Q trunk to core
30	Sales	192.168.30.0/24	L2 access switch, 802.1Q trunk to core
40	IT	192.168.40.0/24	L2 access switch, 802.1Q trunk to core

Component	Specification	Purpose
Core Switch	L3 switch with an SVI (Switched Virtual Interface) per VLAN	Inter-VLAN routing between departments
Router/Firewall	Edge router with NAT and firewall/ACL functionality	Internet access and security policy enforcement between departments and the outside network
Redundancy (optional)	Dual core switches with HSRP/VRRP gateway redundancy	Ensures inter-VLAN routing survives a single core switch failure

5.4 Testing & Validation

- VLAN isolation test: capture traffic on the Finance VLAN and confirm HR broadcast traffic does not appear – verifying broadcast domains are correctly separated.
- Inter-VLAN routing test: ping/traceroute from a Finance PC to an IT server, confirming the core switch correctly routes between VLANs while any configured ACL rules are enforced.
- Gateway failover test (if redundant core switches are used): fail the active core switch and confirm HSRP/VRRP fails over with no extended outage.
- Broadcast-domain comparison: measure broadcast traffic volume per department before/after VLAN segmentation to quantitatively confirm the reduction versus a flat network baseline.

5.5 Innovation / Enhancement Beyond the Base Design

- Implement 802.1X port-based authentication (Network Access Control) on access switches so only authorized, authenticated devices can join each department's VLAN.
- Use RADIUS-based dynamic VLAN assignment, so a user's device is automatically placed into the correct department VLAN based on their login identity rather than which physical port they plug into – useful for hot-desking or shared spaces.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing &	5%	Thorough testing with	Adequate	Minimal	No testing

Validation		real-world scenarios	testing	testing	
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution